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The effectiveness of Transcranial Photobiomodulation therapy (tPBM) on reducing anxiety, depression, and opioid craving in patients undergoing methadone maintenance treatment: a double-blind, randomized, controlled trial

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Abstract

Background The effects of tPBMT are from influencing compounds by triggering specific reactions, which stimulate Adenosine triphosphate biosynthesis and neurogenesis. According to these effects, tPBMT has been applied as a potential treatment for various neural-related diseases. The purpose of this study was to evaluate the impact of tPBMT on reducing anxiety, depression, and opioid craving in patients undergoing MMT.

Methods This randomized controlled clinical trial included two groups of substance-dependent patients undergoing MMT. They were randomly assigned to receive tPBMT or a sham tPBMT. The intervention group received PBM in the form of Light- Emitting Diodes (LEDs) for four minutes of light exposure at 810 nm wavelength producing 250 mW/cm^2 when applied to 4 mm skin depth (totaling 60 J/cm^2) in both forehead locations. The levels of anxiety, depression, and opioid craving were compared between the two groups before and after the intervention, as well as at one-month and three-month follow-up assessments.

Results Both groups consisted of 32 (91.4%) males and 3 (8.6%) females, with the mean age of patients in the intervention group being 37.97 ± 10.58 years, and 39.66 ± 9.94 years in the control group ($P = 0.495$). There were no significant differences between the two groups in terms of depression, anxiety, and opioid craving scale scores before the intervention ($p>0.05$). However, the tPBMT group had statistically significant reductions in their scores compared to the sham tPBMT group. ($p<0.05$).

Conclusion tPBMT led to significant improvements in anxiety, depression, and opioid craving among individuals in MMT, and these improvements were sustained at one month and three months after treatment, indicating a long-lasting positive effect.

Trial registration Trial registration: IRCT code: IRCT20210502051162N1, Approval ID: IR.SBMU.MSP.REC.1400.111, registered on 01.06.2021

Keywords Photobiomodulation, Methadone maintenance treatment, Anxiety, Depression, Opioid craving

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Introduction

Photobiomodulation (PBM) is a non-invasive therapeutic modality introduced by Endre Mester in Hungary around 50 years ago [1]. This technique utilizes red and near-infrared light to stimulate healing, relieve pain, and reduce inflammation [2]. Specifically, PBMT applies targeted light wavelengths to biological tissues, promoting cellular repair and modulating neural activity [3]. tPBM, offers a non-invasive approach to neural modulation. This involves transcranial exposure of the human head to near-infrared (NIR) light, which can penetrate the skin, scalp, and skull and reach the brain. Commonly used wavelengths in tPBM include 660 nm, 800–850 nm, and 1,064–1,070 nm, which activate cellular mechanisms and promote ATP production and local blood oxygenation [4].

Studies suggest that specific parameters can optimize tPBMT efficacy. Wavelengths between 808–835nm, laser devices, higher power densities, and pulsed waves may reduce brain heating and improve therapeutic outcomes [5]. The mechanism of action involves triggering specific reactions, releasing radicals, and altering material properties to stimulate Adenosine triphosphate biosynthesis and neurogenesis [6–8]. As a result, tPBMT has shown promise in treating various neural-related diseases, including traumatic brain injury, ischemic stroke, depression, and anxiety, as well as cognitive disorders like Alzheimer's and Parkinson's [9, 10]. The therapy's impact on brain regions critical to mood regulation is particularly interesting. The prefrontal cortex and precuneus, both part of the default-mode network, are key areas where neuronal effects have been measured through spectroscopy, positron emission tomography, and electroencephalography [11, 12].

Given the promising effects of tPBMT on brain activity and its potential in treating neural-related disorders, its application may also offer therapeutic benefits for individuals with opioid use disorder (OUD), particularly those experiencing anxiety and depression.

Opioid use disorder (OUD) remains a significant global public health challenge often accompanied by psychiatric disorders such as anxiety and depression [13, 14]. MMT is the most widely used approach for managing substance dependence, effectively alleviating withdrawal symptoms, and reducing mortality rates [15–19]. The intricate relationship between substance use, mental health, and addiction underscores the need for comprehensive treatment approaches.

Given the potential safety and promising effects of transcranial photobiomodulation, our study aimed to evaluate its impact on reducing anxiety, depression, and opioid craving in patients undergoing methadone maintenance treatment. We hypothesized that tPBMT could provide a novel therapeutic approach to addressing the complex mental health challenges associated with substance use disorders.

Materials and methods

This study was a parallel, double-blind, randomized, controlled clinical trial involving 70 patients. In addition to the pre-test and post-test assessments, two follow-up evaluations were conducted one month and three months after the intervention.

Participant selection

Inclusion Criteria included Patients aged 18–60 years receiving MMT for at least three months.

Exclusion criteria included concurrent alternative drug addiction treatments (In fact, no other medications aside from methadone were allowed), inconsistent methadone dosage in the preceding three months, history of psychotic or severe psychiatric disorders (By conducting a psychiatric diagnostic interview based on the SCID and by an experienced psychiatrist), active substance abuse within the three months prior to the study, history of violence, suicide attempts, current suicidal ideations, and neurological conditions (e.g., epilepsy, Parkinson's Disease, traumatic brain injury, stroke). These exclusion criteria were implemented to minimize potential confounding factors that could affect treatment outcomes. Pregnant women and individuals with acute or chronic physical illnesses were also excluded. Randomization was performed using a block randomization method by a third-party individual who was blinded to the study protocol. The two groups were matched in terms of demographic characteristics such as sex, age, education level, marital status, and occupation. The study population was predominantly male, reflecting the lower prevalence of methadone use among women [20]. Additionally, the methadone usage amount and duration were balanced between the intervention and control groups. After obtaining written informed consent from the participants, their levels of anxiety, depression, and opioid craving were assessed using standardized scales. Subsequently, they were divided into two groups: 35 patients received tPBMT while 35 were assigned to the control group who received inactive photobiomodulation (device turned off, sham treatment). The intervention group received Photobiomodulation (PBM) in the form of Light- Emitting Diodes (LEDs) as an adjuvant treatment involving four minutes of light exposure at 810 nm wavelength producing 250 mW/cm^2 when applied to 4 mm skin depth (totaling 60 J/cm^2) in both forehead locations. The logic of the optical parameters was as follows: 810 nm wavelength was optimal for light penetration into living tissue due to the lowest amount of absorption by all

three main tissue chromophores, hemoglobin, melanin and water.

In addition, it has been shown that this wavelength is effectively absorbed by mitochondria, which is thought to be responsible for the biological effects of light modulation [21]. The relevant energy density was selected according to other published studies of transcranial lasers for stroke in humans and knowledge about the optical properties of human tissue. The power density (250 mW) was chosen to be safe and prevent skin heating. Both groups attended nine sessions and assessments were conducted immediately after one month and three months following the sessions. The consort flow diagram is shown in Figure 1. Psychological status was evaluated using the Hamilton Depression Rating Scale (HAM-D) with 21 items, Hamilton Anxiety Rating Scale (HAM-A) and Clinical Opioid Withdrawal Scale (COWS), which are well-established, gold standard research tools for assessing depression, anxiety and opioid withdrawal symptoms [22–24]. Side effects of treatment were assessed using open-ended questions and a list of physical and mental symptoms. This study was conducted by the Declaration of Helsinki. Informed consent was obtained from all patients, and the study protocol was ethically approved by the ethics committee at Shahid Beheshti University of Medical Sciences (Approval ID IR.SBMU.MSP.REC.1400.111). Also, the clinical trial was registered under IRCT20210502051162N1 in the Iranian Registry of Clinical Trials (Registration date:2021-08-25). The Kolmogorov-Smirnov test was used to check the normality of the data and Levene's test was used to examine the homogeneity of variance of the data. Statistical analysis was performed using means, standard deviations, numbers, and percentages. Qualitative variables were compared using the chi-square test or Fisher's exact test, while quantitative variables were compared using an independent t-test. A p-value less than 0.05 was considered statistically significant. IBM SPSS Statistics for Windows Version 26.0 was used for data analysis.

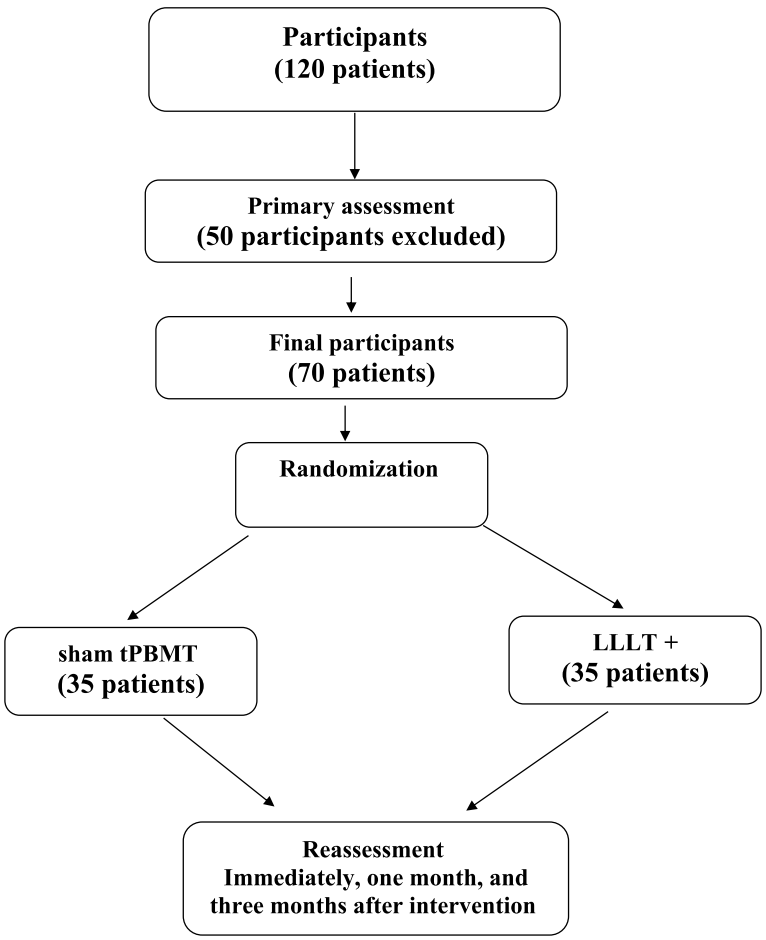


Fig. 1 Consort flow diagram

Results

In this study, both groups consisted of 32 males (91.4%) and 3 females (8.6%). The mean age of patients in the intervention group was 37.97 ± 10.58 years, while it was 39.66 ± 9.94 years in the control group ($P = 0.495$). Table 1 illustrates that there were no significant differences in educational level, marital status, and occupation between the two groups ($p>0.05$). The mean methadone doses in the two groups were 20.30 ± 8.81 cc and 18.40 ± 7.72 cc, respectively, with no significant difference ($P = 0.337$). Additionally, the mean duration of methadone use in the intervention and control groups was 6.01 ± 4.20 years and 4.91 ± 3.62 years, respectively ($P = 0.245$). There were no significant differences between the groups in terms of a history of using other drugs, smoking, neurological and mental diseases, suicide attempts, or use of psychiatric medications ($p>0.05$).

Depression

The HAM-D, used to assess the severity of depression, consists of 17 items, with each item scored on a scale from 0 to 4. This scale was developed by Max Hamilton in 1960 and is widely used in clinical settings. The scale has demonstrated high reliability and validity across various populations [22].

Before the intervention, the mean HAM-D score in both groups was 19.64 ± 3.52 and 19.45 ± 3.11 respectively, indicating no significant difference between the two groups (*P*-value=0.709). However, after the intervention, the sham tPBMT group consistently displayed significantly higher HAM-D scores immediately, one month, and three months later (*P*-value=0.001). At the end of the first and third months, patients in the LLL group had lower depression scores than the control group compared to pre-treatment, with significant effect sizes of 3.79 and 3.75, respectively.

Assessment of depression severity one-month post-intervention revealed that in the sham tPBMT group, 33 individuals (34%) had no depression, while none in the tPBMT group were without depression. Mild depression was observed in 55 individuals (56.7%) in the sham tPBMT group and 17 individuals (23%) in the tPBMT group. Furthermore, moderate depression was present in 9 individuals (9.3%) in the control group and 33 individuals (44.6%) in the tPBMT group. Lastly, severe depression was found in 24 cases (32.4%) in the sham tPBMT group

Table 1 The baseline characteristics of patients in the LLLT and control group

Characteristics	LLLT group	control group	*P*-value
Educational level Num (%)			0.670
Illiterate	1 (2.9%)	2 (5.7%)	
Literacy for reading and writing	7 (17.1%	7 (20%)	
Guidance or high school	15 (42.9%)	11 (31.4%)	
Diploma	11 (31.4%)	10 (28.6%)	
Postgraduate Diploma and above	1 (2.9%)	5 (14.3%)	
Total	35 (100%)	35 (100%)	
Marital status Num (%)			0.672
Single	10 (28.6%)	10 (28.6%)	
Married	20 (57.1%)	23 (65.7%)	
Divorced	2 (5.7%)	1 (2.9%)	
Widow/ widower	3 (8.6%)	1 (2.9%)	
Total	35 (100%)	35 (100%)	
Occupation Num (%)			0.671
Housewife	1 (2.9%)	3 (8.6%)	
Unemployed	4 (11.4%)	3 (8.6%)	
Retired	1 (2.9%)	2 (5.7%)	
Employed	29 (82.9%)	27 (77.1%)	
Total	35 (100%)	35 (100%)	
Mean methadone dose, CC Mean ± SD	20.30 ± 8.81	18.40 ± 7.72	0.337
The mean duration of methadone use, years Mean ± SD	6.01 ± 4.20	4.91 ± 3.61	0.245
other opioids use Num (%)	29 (82.9%)	32 (91.4%)	0.784
History of cigarette smoking Num (%)	30 (85.7%)	29 (82.9%)	0.546
Mean cigarette smoking, years Mean ± SD	15.00 ± 9.27	16.61 ± 11.40	0.556

and none in the tPBMT group (P = 0.001). (Table 2, Figure 2)

Anxiety

For assessing anxiety, the Hamilton Anxiety Rating Scale (HAM-A) was used, which contains 14 items, with scores ranging from 0 to 56, where higher scores indicate greater anxiety. The HAM-A, developed by Max Hamilton in 1959, is a widely recognized tool for assessing anxiety severity in clinical settings. This scale also shows strong validity and reliability [23].

Before the intervention, the average HAM-A scores in both groups were 21.00±5.93 and 20.27±6.67, respectively, with no significant difference between the two groups (P-value = 0.451). However, after the intervention, the sham tPBMT group consistently showed significantly higher HAM-A scores immediately, one month, and three months later (P-value=0.001). At the end of the first and third months, patients in the LLL group had lower depression scores than the control group compared to pre-treatment, with significant effect sizes of 1.48 and 1.51, respectively.

Specifically, one month after the intervention, anxiety was absent in 3 individuals (3.1%) in the sham tPBMT group, while none were without anxiety in the tPMBT group. Mild anxiety was observed in 92 individuals (94.8%) in the sham tPBMT group and 33 individuals (44.6%) in the tPBMT group. Mild to moderate anxiety was reported in 2 individuals (2.1%) in the sham tPBMT group and 29 individuals (39.2%) in the tPBMT group. Finally, moderate to severe anxiety was found in 12 individuals (16.2%) in the sham tPBMT group and none in the Tpbmt group (P-value = 0.001). (Table 2, Figure 2)

Opioid craving scale

For evaluating opioid cravings, the Opioid Craving Scale (OCS) was utilized. The OCS consists of 15 items, with scores ranging from 0 to 60, with higher scores indicating more intense cravings. This scale is specifically designed for assessing cravings in individuals with opioid use disorder and has been widely validated in clinical studies [24].

Before the intervention, the mean OCS scores in both groups were 22.42 ± 3.50 and 22.76 ± 3.31, respectively, showing no significant difference (P-value =0.528). However, post-intervention, the sham tPBMT group consistently exhibited significantly higher OCS scores immediately, one month, and three months later (P-value=0.001). At the end of the intervention, first and third months, patients in the LLL group had lower depression scores than the control group compared to pre-treatment, with significant effect sizes of 2.40, 4.13, and 4.15, respectively. (Table 2, Figure 2).

Discussion

In our assessment, we looked into the therapeutic benefits of PBMT in reducing anxiety, depression, and drug cravings among patients undergoing MMT compared to a control group. Our findings showed that PBMT had significantly improved anxiety and depression scores while reducing drug cravings. Particularly within one month after the therapy, patients who received PBMT showed a more significant reduction in anxiety, depression, and psychological drug craving scores compared to

Table 2 The score of depression, anxiety, and opioid craving of patients in the LLLT and control group before and after intervention

Variables	LLLT group Mean ± SD	control group Mean ± SD	Effect Size	*P*-value
Depression score				
Before intervention	19.64 ± 3.52	19.45 ± 3.11	0.073	0.640
Immediately after intervention	15.06 ± 2.93	17.65 ± 2.67	1.46	0.001
One month after the intervention	9.94 ± 2.48	15.54 ± 2.61	3.79	0.001
Three months after the intervention	9.82 ± 2.52	15.5 ± 2.58	3.79	0.001
Anxiety score				
Before intervention	21.00 ± 5.93	20.27±6.67	0.10	0.451
Immediately after intervention	15.41 ± 4.40	18.09 ± 5.70	0.72	0.001
One month after the intervention	10.33 ± 1.97	17.32 ± 5.48	1.48	0.001
Three months after the intervention	10.07 ± 2.1	17.04 ± 5.55	1.52	0.001
Opioid craving scale				
Before intervention	22.42 ± 3.50	22.76 ± 3.31	0.10	0.528
Immediately after intervention	14.47 ± 2.39	20.82 ± 2.82	2.40	0.001
One month after the intervention	9.01 ± 2.22	19.65 ± 2.78	4.13	0.001
Three months after the intervention	8.84 ± 2.22	19.05 ± 3.77	4.15	0.001

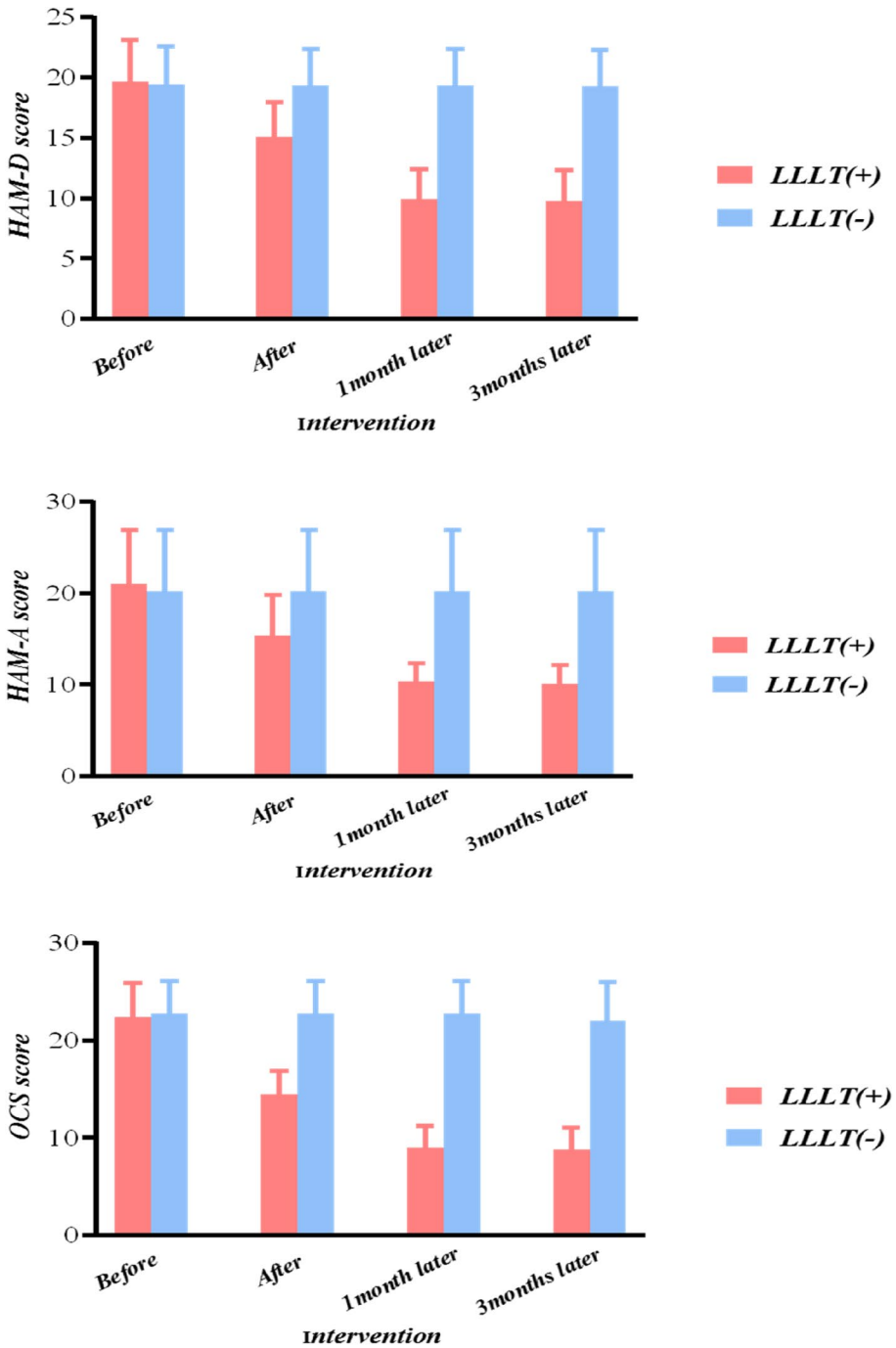


Fig. 2 The score of depression, anxiety, and opioid craving of patients in the LLLT and control group before and after intervention

the control group. These positive effects persisted at both the one- and three-month post-intervention assessments. The increased number of treatment sessions and the specific wavelength used in our study may have contributed to these improvements, along with the lasting impact of laser therapy on brain tissue. Our results align with previous research supporting the effectiveness of PBMT in treating mood disorders, particularly in patients undergoing maintenance treatment. For example, Yang et al. demonstrated that PBMT in animal models with anxiety and depression led to significant improvements in anxiety-like behaviors and depressive symptoms, as well as reductions in neuronal damage, apoptosis, oxidative stress, and inflammation, although memory levels

remained unchanged [11]. Similarly, Kerppers et al. found that PBMT in students with anxiety and depression significantly reduced HADS depression scores compared to a control group [25]. These findings support our study. Alipour et al. observed a significant reduction in Beck depression scores after PBMT treatment [26]. Schiffer et al. also reported a 60% improvement in anxiety and depression scores within two weeks of PBMT treatment, without any side effects. Furthermore, Henderson and Morris noted significant reductions in depression and anxiety scores, along with improvements in cognitive status and quality of life following PBMT treatment [27]. Blanco et al. found a significant improvement in depression scores with transcranial PBMT treatment [28]. Although these studies consistently support the positive effects of PBMT on mood disorders, it is important to consider potential limitations in the existing literature. For example, while many studies have shown positive outcomes, some have noted mixed results, especially regarding the long-term sustainability of these effects. Additionally, variations in treatment protocols, such as different wavelengths and treatment durations, make it difficult to draw broad conclusions about the optimal conditions for PBMT. These inconsistencies underscore the need for further research with more standardized protocols to clarify the ideal parameters for PBMT use. These inconsistencies underscore the need for further research with more standardized protocols to clarify the ideal parameters for PBMT use.

The mechanisms underlying these mood improvements likely involve the modulation of cellular energy production and neurotransmitter activity. Individuals with depression and anxiety, particularly those in recovery from substance use disorders, often exhibit reduced mitochondrial activity and imbalances in neurotransmitters such as serotonin, dopamine, and noradrenaline. PBMT has been shown to enhance mitochondrial respiration, promote ATP synthesis, increase reactive oxygen species expression, stimulate nerve cell proliferation and migration, and modulate these key neurotransmitters, as confirmed by various studies [8, 29]. These biological changes likely contribute to the observed improvements in mood among patients treated with PBMT.

While our study suggests that PBMT is a promising therapeutic option for individuals with mood disorders and substance use issues, there are several limitations. First, data collection occurred at a single substance abuse center, which may limit the generalizability of our findings. Conducting multicenter studies would help validate the results and increase their reliability. Additionally, the sample size in our study was relatively small, which may limit the statistical power and generalizability of the findings. Moreover, we were unable to conduct long-term follow-up assessments to evaluate the sustainability of the observed improvements, a crucial aspect for understanding the lasting impact of PBMT. Therefore, we recommend further studies with larger sample sizes, multicenter involvement, extended treatment durations, and long-term follow-up to provide a more comprehensive understanding of the benefits of PBMT.

Another key limitation of this study is the suboptimal sham tPBMT condition used. A stronger control condition would involve actual stimulation to a region that is not thought to be involved in the therapeutic effects. Future studies incorporating a more rigorous control condition, such as stimulation to an irrelevant brain region, would provide more robust evidence regarding the specificity of the effects and the underlying mechanisms.

Conclusion

Transcranial PBMT showed significant improvements in anxiety, depression, and OCS among individuals undergoing MMT. These improvements were sustained at one month and three months after treatment, indicating a long-lasting positive effect. PBMT's ability to alleviate mood disorders and reduce cravings demonstrates its potential as a valuable therapeutic approach for individuals in recovery from substance abuse. Given the substantial impact of PBMT on both mental health and substance dependence, it represents an important treatment option for patients undergoing MMT, offering a safe, non-invasive alternative that could enhance their recovery process and quality of life.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12888-025-06555-3.

Supplementary Material 1.

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Authors' contributions
H H, N S, F T, and A Kh conceived the presented idea. All authors (H H, N S, F T, Sh E, and A Kh) participated in the study design, the conception of research questions, and the drafting of the paper. H H, Sh E, N S, and A Kh led the work and wrote the first draft of the paper. All authors reviewed the manuscript.

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Data availability
While no specific data sets are reported in this manuscript, any relevant data supporting the findings of this study can be made available upon reasonable request. Please contact the corresponding author for further information regarding data availability.

Declarations

Ethics approval and consent to participate

The Declaration of Helsinki conducted this study. Informed consent was obtained from all patients, and the ethics committee ethically approved the study protocol at Shahid Beheshti University of Medical Sciences (Approval ID IR.SBMU.MSP.REC.1400.111). Also, the clinical trial was registered under IRCT20210502051162N1 in the Iranian Registry of Clinical Trials (Registration date:2021-08-25).

Consent for publication

Not Applicable.

Competing interests

The authors declare no competing interests.

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